

SmartTAR STAR v1.2.0

Release Summary - Responsive Analysis Engine, Safer Staging, and STAR Block Optimizations

Overview

SmartTAR STAR v1.2.0 is a stability, performance, and architecture-focused release. SmartTAR remains a smart PowerShell wrapper and STAR container orchestrator built on top of Windows tar.exe / bsdtar, not a custom compression engine.

The release improves content-aware archive planning, internal block layout, verification reliability, and GUI responsiveness while keeping the project dependency-light and Windows-native.

Main Improvements

Area	Summary
CPU-aware analysis	Parallel content analysis via PowerShell RunspacePool with safe worker scaling: 1 / 2 / 4 workers depending on logical CPU count.
Smart profile	Full content analysis with max-compression block mapping: structure/text/unknown/binary → XZ9, archives → STORE.
Structure block	Directory structure block is now compressed as structure.tar.xz when XZ9 is available, with STORE fallback.
Staging resilience	Hardlink staging remains preferred; copy-staging fallback improves cross-volume behavior before chunked fallback is used.
Cleaner profiles	User-facing profiles are now Balanced, Smart, Solid, and Store. Older Hybrid / SmartXZ naming was removed.
Stable output	Progress messages and final diagnostics were simplified for a cleaner stable release report.

Current Smart Profile Block Strategy

```
structure → XZ9
text      → XZ9
unknown   → XZ9
binary    → XZ9
archives  → STORE
```

This strategy keeps already-compressed or archive-like data stored without unnecessary recompression, while compressible data is grouped and compressed using XZ9.

Validation Snapshot

Profile / Check	Result
Smart - max compression	OK
Balanced - mixed blocks	OK
Solid - single block	OK
Store - no compression	OK
Verify	OK

Example Smart Profile Result

```
Source size: 415.37 MB
Archive size: 140.79 MB
Ratio: 33.89 %
Saved: 66.11 %
Verification: OK
```

Internal STAR Layout

A typical Smart profile archive now uses an internal layout similar to:

```
manifest.json
blocks/
  000001_structure.tar.xz
  000002_text.tar.xz
  000003_unknown.tar.xz
  000004_archives.tar
  000005_binary.tar.xz
```

Design Philosophy

- SmartTAR is a wrapper and orchestrator, not a custom compression algorithm.
- The STAR container adds manifest metadata, block grouping, hashing, diagnostics, verification, and salvage-friendly structure.
- No external compressor dependencies are required. The project remains based on Windows tar.exe / bsdtar.
- The v1.2.0 release is considered the stable v1.2 baseline.

Release Highlights

- CPU-aware parallel content analysis.
- Compressed structure block: approximately 120 KB reduced to approximately 2.5 KB during validation.
- Cleaner compression profiles and stable reports.
- Verified STAR archive integrity across all main profiles.